

Data structures

Q1.

Carefully analyze the algorithm and solve the questions that follows.

```
S1   a ← sign(X) * sign(Y);
S2   X ← abs(X)
S3   Y ← abs(Y); (make X and Y positive)
S4   if      n = 1 then
S5       if      (X = 1) and (Y = 1) then
S6           return(S)
           Else
S7           return(0)
           Else begin
S8   A ← left n/2 bits of X;
S9   B ← Right n/2 bits of X;
S10  C ← Left n/2 bits of Y;
S11  D ← Right n/2 bits of Y;
S12  M1 ← Mult (A, C, n/2);
S13  M2 ← (A-B, D-C, n/2);
S14  M3 ← Mult(B, D, n/2);
S15  return  ( S*      (M1*2n + (M1 + M2 + M3) + 2n/2 + M2.....)      )
END
```

END (MULT)

- What is the underlying algorithm?
- Use the underlying algorithm to solve the product of 1011 and 1111
- Distinguish between the worst – case and the average case remaining
affect the running time of an algorithm.

Q2.

A stack is implemented as an array Stk of size n, and indexed from 1. The stack pointer, T points to the topmost item on the stack. An algorithm that pushes m integer values and the value m, one after the other, onto a stack, is used to build a stack that has several groups of integers, such that the number of items in each group is pushed onto the stack after members of the group have each been pushed onto the stack.

Write an algorithm that will empty the stack and display the **total number** of **groups** of integers that were held on the stack.

Q3

A geometric figure has n sizes and they are $Q_1, Q_2, Q_3, \dots, Q_n$ as input such that

$$A_1 = Q_1$$

$$A_2 = Q_1 + Q_2$$

$$A_3 = Q_1 + Q_2 + Q_3$$

.

.

.

$$A_n = Q_1 + Q_2 + Q_3 + \dots + Q_n$$

It is then transformed to produce an output $A^1_1, A^1_2, A^1_3, \dots, A^1_n$. What is the underlying algorithm.

Q4.

If the number of statements in sequence P is constant and independent of n , determine the number of times P is executed, and the worst time complexity of the following algorithm.

```
for i = 1 to n step 1
    for j = i to n step 1
        P
    end for
end for
```

Q5.

Show that $O(n) = O(n + k)$, where K is any positive constant.

Q6.

What will be the time complexity of an algorithm being a linear search if the desired element is the first element of the array?

Q7

- a. Let T be a pointer to a linked list. If $T = 0$ [No list] it means the list has no node. Let X be a pointer to any arbitrary node in the list T. Write an algorithm to insert a node with the data field "OAT" following the pointer at X.
- b. Assuming each node has two fields, data field and link field. Write an algorithm to create a list with 2 nodes whose data fields are said to be of the values "Mat" and "PAT" respectively. T is the pointer to the very first node of the list.

Q8.

List and explain the factors that affect the running time of algorithms.

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tg.

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ONE HOUR

INDEX NUMBER _____

Answer ALL questions by circling on the question paper as well as shading the letter of the best answer on the scannable sheet, and where applicable, using the spaces provided.

1. A file is a collection of similar records organized _____.
☒ a. externally
☐ b. internally
☐ c. randomly
☐ d. none of the above
2. A main program principally calls a number of procedures or functions. Such a program is called a _____ program.
☒ a. driver
☐ b. parent
☐ c. root
☐ d. test
3. Indicate the odd one out.
☒ a. bits
☐ b. colours
☐ c. traffic signals
☐ d. whole numbers
4. Indicate the odd one out.
☐ a. bits
☐ b. data
☐ c. program
☒ d. storage
5. To access an item, in computing, means to _____ the item.
☐ a. insert or delete
☐ b. search for
☐ c. create
☐ d. store or retrieve
6. The appropriate data structures to use when insertions and deletions are frequent are _____.
☐ a. stacks and queues
☐ b. arrays and records
☐ c. arrays and queues
☐ d. records and stacks
7. When you type at the computer's keyboard the corresponding characters are

- stored in an area that functions as a _____.
- table
 - file
 - ☒ queue
 - stack
8. Breaking a large problem into several smaller problems and solving them one at a time is called _____.
- top-down development
 - stepwise refinement
 - ☒ divide and conquer approach
 - bottom-up development
9. What feature of C++ facilitates the top-down approach when coding the solution of a problem?
- ☒ functions
 - structure charts
 - comments and documentation
 - bottom-up
10. Replacing a step with more detailed steps in the solution of a problem is referred to as _____.
- divide and conquer
 - stepwise refinement
 - repetition
 - ☒ top-down approach
11. The display of the hierarchical organization of the sub problems of a problem is called a _____.
- flowchart
 - procedure chart
 - ☒ structure chart
 - systems chart
12. A digital computer holds physical encodings of symbolic information. The operation that retains the encoding but changes the manner in which the individual elements of the encoding are represented is termed _____.
- ☒ input or output
 - processing
 - storage
 - transmission
13. A pointer variable is used to indicate to _____.
- indicate whether or not an element exists
 - ☒ store the address of a node
 - store the address of each field of a record
 - store the address of each entry in an array
14. Each node of a linear data structure in contiguous memory locations takes 50 bytes of storage. The first node has address 500. What is the address of the 101st node in the structure?
- ☒ 5450
 - 5500
 - 5550
 - 5600
15. Using the data of Question 14, the _____st node is at address 10000.
- 181
 - 186

- c. 191
d. 196
16. A computer that takes a microsecond to list a subset, will list all the subsets of a set of 24 members in approximately _____ seconds.
a. 5
b. 9
c. 17
d. 33
e. none of the above
17. For the functions $g(n) = n^3$ and $h(n) = 3^n$, $g(n)$ dominates $h(n)$ in terms of execution time.
a. True
b. False
18. The relationships among the nodes in a network data structure are _____.
a. one-to-one
b. one-to-many
c. many-to-one
d. many-to-many
e. none of the above
19. The relationships among the nodes in a tree data structure are _____.
a. one-to-one
b. one-to-many
c. many-to-one
d. many-to-many
e. none of the above
20. The information for processing in on a stack is referenced by _____.
a. address
b. order of placement
c. name
d. none of the above
21. A pointer variable is used to _____.
a. indicate whether or not an element exists
b. store the address of a node
c. store the address of each field of a record
d. store the address of each entry in an array
e. none of the above
22. In terms of method of access, indicate the odd one out.
a. array
b. list
c. stack
d. queue
e. none of the above
24. In terms of the notion of data, indicate the odd one out.
a. array
b. list
c. stack
d. queue
e. none of the above
25. When you type at the computer's keyboard the corresponding characters are stored in an area that functions as a _____.

- a. table
 - b. file
 - c. stack
 - d. list
 - e. none of the above
26. The display of the hierarchical organization of the sub tasks of a problem is called a _____.
- a. flowchart
 - b. procedure chart
 - c. project chart
 - d. systems chart
 - e. none of the above
27. Classify the big O measure of $O(n^3)$.
- a. constant
 - b. polynomial
 - c. logarithmic
 - d. exponential
 - e. none of the above
28. Classify the big O measure of $O(3^n)$.
- a. constant
 - b. polynomial
 - c. logarithmic
 - d. exponential
 - e. none of the above
29. Determine the order of $g(n) = (n * \log n) / (2 + n)$
- a. $O(1)$
 - b. $O(n)$
 - c. $O(\log n)$
 - d. $O(n \log n)$
 - e. $O(2^n)$
30. Determine the order of $g(n) = \log n + 2 * n$
- a. $O(1)$
 - b. $O(n)$
 - c. $O(\log n)$
 - d. $O(n \log n)$
 - e. $O(2^n)$

$$\frac{n \log n}{2 + n}$$

$$\log n + 2n$$